

Week 6 Project Summary

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1. Planned for Week 6

Deepen and validate the strongest findings from Week 5 (previous no-show history and distance to clinic), rather than repeating the Week 5 EDA, and use the results to complete a meaningful cross-track integration with the Data Science track.

2. What Was Completed

Validated the previous-no-show finding across all six age groups, validated the distance finding across all four appointment types, tested the combined effect of both variables together, validated the core No-Show Rate KPI's stability across segments, and produced a ranked, evidence-based feature recommendation for the Data Science track's prediction model.

3. What Was Improved From Week 5

Week 5 treated previous no-show history and distance as separate, independent findings. Week 6 revealed they interact: distance strongly affects patients with low-to-moderate prior no-show history, but has little additional effect on patients who already have a high prior no-show count, since their risk is already near-ceiling. This is a meaningful refinement, not a repeat, of the Week 5 conclusions.

4. What Was Integrated

Delivered a formal, ranked feature recommendation (previous_no_shows as highest priority, a distance x prior-history interaction feature as second priority, appointment_type as a supporting feature, and waiting_time_minutes excluded) intended to directly inform the Data Science track's feature selection process for its no-show prediction model.

5. Track(s) Collaborated With

Data Science.

6. What Was Exchanged

Provided: a validated, ranked feature list with supporting evidence from segment-level analysis. Received (from the Week 5 Data Science problem definition): confirmation that the model is a binary classifier predicting No-Show vs. Attended, which shaped how the feature recommendations were framed.

7. What Changed As a Result

The candidate feature list for the prediction model moved from an assumed, untested list to a ranked, evidence-based one, with a specific recommendation to test an interaction feature rather than treating distance and prior history as fully independent inputs.

8. Key Findings

- Previous no-show history is a consistent, strong predictor across every age group tested, with no exceptions.
- Distance to the clinic matters most for patients with low-to-moderate prior no-show history; its effect weakens once prior history is already high.
- Diagnostic Test appointments show unusually high distance sensitivity compared to other appointment types.
- The core 48.5% No-Show Rate KPI is stable and representative across appointment types and age groups.

9. Major Challenges

Identifying the correct way to test two variables together (previous no-shows and distance) without simply assuming an additive relationship — this required building a combined segment breakdown rather than two separate charts, to reveal the interaction effect.

10. Important Decisions Made

Chose to recommend a distance x prior-history interaction feature to the Data Science track, rather than recommending both variables as independent inputs, based directly on the Week 6 combined-effect finding.

11. Remaining Issues

Some segment combinations (e.g., 3+ previous no-shows at longer distances) have smaller sample sizes, which should be monitored for reliability as the Data Science track builds and tests its model.

12. Contribution to the Overall HealthConnect Project

This week's work moved the Data Analytics track's contribution beyond descriptive findings into a validated, decision-ready input for the modelling stage of the project, directly supporting the Data Science track's Week 6 model improvement work.

13. Proposed Focus for Week 7

Support testing of whether the recommended interaction feature improves the Data Science track's model performance compared to the Week 5 baseline, and further investigate the Diagnostic Test distance-sensitivity finding.